TECHNICAL REPORT — CARTOGRAPHY & STATE OF PLAY

Inter-agent information exchange

EFE’s four coordination layers, their **real** maturity, and the Couche 3 M1 work: serving **real** efe-core participants over gRPC, under *fail-closed* guards and without over-selling.

Program	EFE → Kovant · proof dossier (the “dowry”)
Scope	efe Rust monorepo · HEAD 665ead0 (+ working tree)
Date	July 14, 2026
Author	R. Pekar — with Claude Code (Opus 4.8)
Method	multi-agent code cartography + honesty review; file:line citations
Cardinal rule	NEVER OVER-SELL — every claim is labelled with its verified maturity

Honesty warning (read this first)

This report systematically distinguishes **what is served by default**, **what is real and tested but disabled by default**, and **what is a stub / a canned response**. The system exchanges information between agents in **four** genuinely different ways, spread across **four independent binaries** — and **only one** is on by default. No robust aggregation (weighted vote, median, outlier rejection, Byzantine tolerance) is served anywhere; no non-expansiveness is *proven* (only *falsifiable* through a probe that measures a lower bound). Read maturity **per path**, never as aggregate capability.

1 Executive summary

EFE makes “agents communicate” in four genuinely distinct ways; only one is on by default. This report maps them honestly, then details the only project **we** ran on this axis during this arc — Couche 3 M1, which made a real efe-core participant *servable* as the meta-swarm’s exchange operator, every step locked *fail-closed* and every over-sell retracted.

(A) Layer 1 — the swarm’s latent blackboard. Inside a swarm, the “nano-agents” are rows of a shared latent tensor (bf16); they exchange nothing but their vectors, through a CPU softmax-attention fusion that pulls them toward a centroid (a Fréchet mean). It is real, deterministic, unit-tested, and **served by default** in efe-server (`EFE_BACKEND=mvp`). It is the most mature layer — but it is mean-field matrix math over a shared slab, **not** autonomous agents negotiating messages (an agent is a tensor row; its voice is a vector; the exchange is a *matmul*).

(B) Layer 2 — the multi-LLM frontier panel. A Claude / GPT / Gemini panel genuinely fans a prompt out to N models and exchanges text — but only the *debate* mission passes real arguments; *vote / relay / specialist / cascade* fuse via plurality / Borda / Condorcet, under a re-verifiable provenance receipt. This layer is now **served but off by default**: only in the separate efe-swarm-gateway binary, behind a `frontier-llm` compile feature (default off), a runtime flag and a *fail-closed* boot triad. The efe-server product gateway does not depend on it at all.

(C) Layer 3 — the coupled meta-swarm over gRPC. The controller couples participants by circulating a meta-consensus vector as the *pull* at every step. The transport, mTLS, the coupled iteration, the cancellation/deadline caps and the *fail-closed* gates are real and served — but the **default** participant is a synthetic mock $\Phi(x)=\kappa\cdot x$, aggregated by a plain arithmetic mean (`UniformMetaOp`) with **no robustness whatsoever**. The real efe-core participant, the empirical Lipschitz probe (H1) and the **measured** bounds (not proven: $\hat{L}\approx 0.37$ fixed / 0.69 dynamic, refuting the analytic certificate $\beta_1\approx 0.17$) all live behind the `core-integration` feature, disabled by default. **This is where all of our work lives.**

(D) Layer 4 — the N21 agentic blackboard. A fourth substrate, the most *literally* message-passing one: a blackboard (a versioned JSON-pointer store) + an *EventBus* (`tokio::broadcast`) + a DAG orchestrator + an agentic bridge that calls the **real** Claude (Anthropic Messages API, a two-turn *tool_use* protocol), served by its own efe-gateway binary. But **by default**, without an API key, the endpoint returns a canned response; the *EventBus* has no runtime subscriber; and the served DAG never seeds the input text, so the produced *ContextBundle* is essentially empty.

> **The one sentence to remember**
>
> The **only** form of inter-agent exchange on by default is Layer 1’s latent attention fusion, in efe-server. Every cross-model or cross-process “agent coordination” story is real and tested, but **off by default** or **mock by default**. No robust aggregation is served anywhere (arithmetic mean only); no non-expansiveness is proven (only falsifiable); and the four binaries are **independent, with no shared data path at runtime**.

2 Framing: what does “exchanging information between agents” mean?

The word “agent” covers four very different scales here, and the nature of the “information exchanged” changes radically from one to the next. Conflating those scales is the first source of over-selling; separating them is the precondition of an honest narrative.

- **Nano-agent** (Layer 1) — a *row* of a latent tensor. Its "message" is a vector of $\mathbb{R}^{16}$; "exchanging" = a softmax attention over the whole population, then a fold toward the centroid.
- **Frontier model** (Layer 2) — an LLM (Claude / GPT / Gemini). Its message is *text*; "exchanging" = fanning a prompt out, collecting the answers, and fusing them by vote/Borda/Condorcet (or, in *debate*, re-injecting the opponent's argument into the next turn).
- **Meta-swarm participant** (Layer 3) — an entire swarm seen as an operator Φ. Its message is a *state vector* carried over gRPC; "exchanging" = a coupled fixed point where the controller sends the meta-consensus back as the pull.
- **DAG SwarmAgent** (Layer 4) — a node of an orchestrated graph. Its message is a *JSON cell* written to a shared blackboard (+ an event published on a bus); "exchanging" = reading/writing versioned JSON pointers.

One structural fact, to state up front: these four layers live in **four distinct binaries** (efe-server, efe-swarm-gateway, efe-meta-swarm-server, efe-gateway) that **share no data path at runtime**. The only inter-binary link found in the code is a metrics *type* (`SubstrateMetrics`) reused by efe-swarm-gateway for its `/metrics` endpoint — not a blackboard instance, not a bus, not a shared swarm. Any prose suggesting that "the binaries coordinate through a shared blackboard" would be false.

3 Panorama of the four layers

The following table is the maturity ledger, to be read before any capability claim. The "default status" column is decisive: it says what actually runs in a binary compiled and launched without flags or keys.

Layer	"Agent" =	What is exchanged	Binary / transport	Default status
1 · Latent blackboard	tensor row	latent vector $\mathbb{R}^{16}$ via softmax attention → centroid	efe-server (in-proc)	**SERVED**
2 · Frontier panel	LLM (Claude /GPT/Gemini)	text + provenance receipt (vote/Borda/Condorcet, debate)	efe-swarm-gateway (HTTP)	**OFF** (feature+flag)
3 · gRPC meta-swarm	swarm = operator Φ	state vector (coupled pull) over mTLS	efe-meta-swarm-server (gRPC)	**MOCK** κ·x
4 · Agentic blackboard	DAG node	versioned JSON cell + event; Claude tool_use	efe-gateway (HTTP)	**CANNED** without a key

"SERVED" = active in a default binary. "OFF" = real + tested, but behind a compile feature and a runtime flag. "MOCK" = the transport is served, but the default computation is a synthetic stand-in. "CANNED" = the endpoint answers with hard-coded text until an API key is provided.

Two honesty corollaries follow from the table. First, maturity **decreases** as the "agent" becomes more autonomous: the most mature layer (1) is also the least "agentic" (a *matmul*), and the most literally agentic layer (4) is the one whose default answer is canned. Second, **no** layer serves robust aggregation today: wherever agent voices are fused on the served path, it is a mean (arithmetic in Layer 3, plurality/Borda in Layer 2) with no outlier rejection and no learned weighting.

4 Layer 1 — the swarm's latent blackboard (nano-agents)

Status: **PROD · served** · the most mature layer.

The substrate is the **Epistemic Latent Space** (`efe-core/src/els/mod.rs`): four arena-backed SoA tensors — the latent `[N,D]` in bf16 (one row $h_i \in \mathbb{R}^D$ per nano-agent, the exchange medium), an activity mask, the anchors, and the *problem embedding*. A nano-agent emits neither token nor message: its whole state — and its whole "utterance" — is its latent row. At genesis (`DefaultSpawner`), each row receives h_i = prior[template$_i$] + init_sigma·N(0,1;seed): a template prior (its "persona") plus deterministic per-spec noise. That is where each agent's initial opinion enters the shared latent.

The exchange itself is the **fusion** (`MvpCpuFusion`, `fusion/mvp_cpu.rs`). Each round: attention scores $s_{ij}=(h_i \cdot h_j)/\sqrt{D}/T$ over the whole population, a row-softmax (max-shift stabilised), then the aggregate new[i]=Σ_j softmax$_{ij}$·old[j] into a double-buffered scratch (pointer swap → O(1) memory cost independent of round count); finally a β-mix toward the *problem embedding*, a projection onto the alignment cone, and a clip. This dense N×N attention **is** the inter-agent information exchange: each agent reads every other agent's row and rewrites its own as a consensus-weighted blend. The loop is driven by the `SwarmController` until the consensus detector fires *Converged* (with a `min_rounds=4` floor in Mode-1). The swarm's collective answer is a centroid collapse (`WeightedMeanExtractor`, a uniform arithmetic mean over the active rows), extracted by the `FusionAdapter` and published for the dissolver.

This path is **genuinely served**: efe-server boots in Mode-1 (`EFE_BACKEND=mvp` by default), builds a real `SwarmController` and runs the genesis → fusion → centroid chain. It is, without exaggeration, the most solid part of the whole inter-agent story; understating it would be as dishonest as inflating the stubs.

> **What Layer 1 is NOT (guardrails against over-selling)**
>
> • **No agent negotiation**: it is a mean-field fusion (a matmul), not debating agents. • **No self-healing**: mode-collapse recovery (`respawn_diversify`, `inject_noise`) is a logged *no-op* — honest only because the Mode-1 wiring disables recovery. • **No Langevin/SDE**: the served round is attention + β-mix + projection + clip, with no per-round Gaussian noise (the word "Langevin" only appears in a trait docstring). • **CPU only**: the CUDA kernels are "Sprint 3+/7" notes — no GPU acceleration is served. • **Uniform mean**: outlier-robust centroid weighting (`DistanceInverse`) is "Sprint 8+", so a few outlier rows can drag the centroid.

Finally, a state note: the "pure-Φ by construction" fix (carrying the centroid on the ELS via `els.dissolution_centroid` / `ElsCarriedDissolver`, to make Φ pure under concurrent `phi_once`) is **uncommitted** in the working tree (diff: els +49, controller +23, mode1 +187, test +94) and only affects the meta-swarm's latent controller, never the served path (byte-identical). A clean checkout at HEAD would not have it. Report it as *in flux*, not as a served Layer-1 guarantee.

5 Layer 2 — the multi-LLM frontier panel (Claude / GPT / Gemini)

Status: **real · off by default** · served *only* in efe-swarm-gateway, feature+flag, absent from the product server.

The physical transport is the **MultiLlmBridge** (`efe-coordinator/src/llm/bridge.rs`): a parallel fan-out to N providers (`FuturesUnordered`), collecting one response per provider. The backends are **real** reqwest HTTP clients to `api.anthropic.com/v1/messages`, `api.openai.com/v1/chat/completions` and `generativelanguage.googleapis.com` (keys via `from_env`). Five missions embody the exchange topologies: *vote* (N-way fan-out then aggregation), *debate* (pro/contra LLMs alternating, where the turn-N prompt embeds the opponent's turn-N−1 argument — the only genuine transcript exchange), *relay* (a plan→code→synthesis chain), *specialist* (decompose→N experts→aggregate), *cascade* (drafter→verifier→validator).

Answer fusion goes through a reformulator (each raw output becomes a *CanonicalClaim*) then a plurality, Borda or Condorcet tally. An honesty fact: Borda's "inter-model ranking" is a **heuristic** (each LLM ranks itself first, then orders the rest by Jaccard similarity — the comment says "Heuristique simple"): the models do not actually rank each other; only *debate* performs real argument passing. Weighting is uniform — **no** weighted vote and **no** outlier rejection.

Since the COUCHE2 arc (commits f71f5b8 / c7ed62d / f8e70a5), this layer has a **served surface**: `POST /v1/llm/{vote,relay,debate,specialist,cascade}` and `GET /v1/llm/audit/:id` in efe-swarm-gateway, each mission under a provenance scope that seals a **re-verifiable receipt**. The causal graph is built from the **declared** topology (not the observed records), so a missing egress call becomes an unlinked node → `verified=false`: that closes the "empty receipt re-proves fully-valid" hole. A *fail-closed* sovereignty gate (`egress_gate`) refuses (403) any LLM outside the allowlist or any missing tenant policy **before** a single call.

> **The double lock — and why "in production" would be false**
> "Served" here is doubly gated and absent from the product server: (1) the `frontier-llm` compile feature, whose default is empty (off); (2) runtime `EFE_LLM_FRONTIER_ENABLED=true` exactly, plus a *fail-closed* boot triad (HMAC key ≥ 32 B, OIDC public key aud=efe-llm, a durable Postgres store) without which boot fails. And efe-server (the product server) has **zero** dependency on this stack. Writing "the platform coordinates Claude/Gemini/GPT in production" would be false for the default build.

- **Silent-drop residue**: the *vote* mission still calls `call_many_ok` (`vote.rs:57`) — a provider failing at the bridge is dropped from the vote's own logic (the provenance decorator still records an *Errored* node for calls that reached the provider, so the receipt is not blind).
- **Integrity ≠ correctness**: the receipt proves *what happened* (which models, which hashes, which vote), not that the answer is right. The distinction is stamped in the code (`HeldStrengthView`, rungs R1..R4).
- **Duplicated egress guard**: the locality classifier (`is_local_endpoint/host_of`) exists as two hand-synced copies (provenance-capture vs efe-swarm-sdk/backend); a historical SSRF drift is noted and re-synced in comments — harden both together (a maintenance hazard, not a current bug).
- **efe-llm-router**: explicitly **rejected** as the coordination carrier (a Postgres audit that is not offline re-provable, temperature=0 killing diversity); frozen, not served.

6 Layer 3 — the coupled meta-swarm over gRPC (our project)

Status: transport **PROD · served** · real participant **real · off by default** · default = **mock only** (κ·x).

This is where **all of our work** concentrates. The wire contract (`protos/efe/v1/meta_swarm/`) defines the exchange: controller-side `Iterate/StepOnce/Reset/ReloadConfig/Health/Info`, participant-side `Step/State/Capabilities/Health`. State on the wire is a dense `TensorF32`; `StepRequest.pulled_from_meta` is the vector exchanged at every step. The engine is a **coupled iteration** (`CoupledIteration`): a fixed point where the controller fuses the participants' votes into a meta-consensus then redistributes it as the pull, per $x_{t+1} = \mathrm{mix}(x_t, \Phi(x_t), \mathrm{pull}; \alpha, \gamma)$. The transport (`RemoteParticipant` → `Step` RPC over mTLS), cancellation, deadline caps: all real and served.

But the participant **served by default** is the `MockParticipant` — a synthetic $\Phi(x)=\kappa\cdot x$, non-expansive by construction — and the aggregator is `UniformMetaOp` (an arithmetic mean, Lipschitz bound exactly

1). It is the **only** MetaOperator ever instantiated anywhere (builder default *and* server hardwire, `iterate.rs:701`). The alternative `AttentionMetaOp` (softmax) is a stub explicitly "NOT validated against the convergence theorem", behind an off feature, whose bound $1+2\|q\|/\tau$ can **exceed 1** (expansive). There is **no** outlier rejection, **no** median/trimmed scheme, **no** learned weighting anywhere.

6.1 · The real efe-core participant (real, but opt-in)

The real participant maps the meta state x → an efe-core *problem embedding*, runs one Mode-1 numerical fusion round (`build_mvp_controller` / `build_fixed_topology_controller`), and returns Φ(x). These are **real efe-core latent swarms** — **not** SLMs: no model weights, no tokenizer on the step path; the exchange is a centroid vector of $\mathbb{R}^{16}$, not model inference. The `core-integration` feature is **disabled by default** in both crates; on a default build, an efe-core manifest row **fails construction** ("built without core-integration") — fail-closed, never a silent fallback to the mock.

6.2 · The H1 hypothesis and its probe — a falsifier, not a proof

Convergence (Krasnoselskii–Mann) rests on hypothesis H1: Φ non-expansive ($L_\Phi \le 1$). The real threat is the **discontinuity** of genesis routing (the class of x → an **integer** number of agents → a staircase topology → Φ discontinuous at the boundaries). The empirical Lipschitz probe (`lipschitz.rs`) measures a **lower** bound Ł by adversarial black-box sampling with bisection toward the discontinuities. It is a **falsifier**: it can **reject** H1 (Ł ≥ 1−margin ⇒ refusal), never **certify** it. Every label reads "non-expansiveness property-tested, NOT certified".

> **The analytic certificate $\beta_1 \approx 0.17$ is REFUTED by measurement**
>
> The closed-form intra-cell bound $L_\Phi \le \beta_1 \approx 0.17$ is **minted nowhere** as a certificate. The M1.c probe measures Ł ≈ **0.366** on the fixed operator and Ł ≈ **0.691** on the dynamic one — both < 1 (H1 holds, property-tested) but both **far above** 0.17: direct evidence of a continuous x-sensitivity beyond *beta_mix*. The only honest H1 evidence is the **measured** bound.

An honesty point we corrected **ourselves** this arc (finding #2): the 0.366 vs 0.691 gap is **confounded**, not causal. `GenesisEngine::spawn` ignores `spec.population` and the dynamic path uses `HeuristicEstimator` (`min_agents=128`): the fixed operator runs 4 pure *na-sem* agents while the dynamic one hatches ≥128 mixed agents. Population size (~32×) **and** template mix change together with routing fixity; the claim "fixing the topology divides the sensitivity by ~1.9" was **retracted**. What survives (a single-operator, non-comparative statement): the fixed operator alone measures 0.366 ≫ 0.17, which refutes the certificate.

6.3 · The fail-closed gates actually wired

- **h1_boot_gate** (`participant/local.rs:343`) — runs the probe **before** the participant binds its port; Ł ≥ 1−margin or inconclusive ⇒ refuses to serve. The only real non-expansiveness gate in the whole layer (at boot, for efe-core engines).
- **require_convergence** (`iterate.rs:211`) — when set, a non-converged Iterate returns gRPC `ABORTED` instead of an untrustworthy "consensus". A response-level gate (default off; StepOnce exempt by design — a single step has no convergence verdict).
- **config_hash on the wire** (`messages.proto`, field 8) — efe-core rows carry a `controller` block whose canonical SHA-256 is advertised; presence ⟺ real engine; absence **must** be read as "unverified / mock" (an old binary prunes the unknown field → fail-closed).
- **fixed topology** (Palier-2, `genesis/fixed.rs`) — makes the population x-independent, so Φ is continuous by construction (removes the routing discontinuity, the one H1 killer). Default topology for efe-core rows; *dynamic* is an explicit opt-out, visible on the wire, logged at boot.

> **What remains a stub in Layer 3 (do not inflate)**
>
> **Wire certificates** are stubs: `RemoteParticipant::verify_non_expansive` always returns `Stub`, so the per-Iterate H1 pre-check is a *no-op* on the served path (lenient mode — strict mode would reject every Stub and brick serving). H1 is **neither exchanged nor checked at runtime** between agents; it is only checked at participant **boot**, and only for efe-core engines (themselves off by default). Signed wire certificates (Ed25519 `LocalSignature`) are milestone M4, **not built**. The served *lifts/pulls* are identity (no learned cross-space projection). Only the `BoxSafety` geometry is wired (Poly/Hull rejected at build).

7 Layer 4 — the N21 agentic blackboard (efe-gateway)

Status: substrate **PROD · served** · Claude bridge **real · off by default** · response **mock only** (canned) without a key.

This is the most **literally** message-passing layer, served by its own axum binary efe-gateway (distinct from the other three). The blackboard (`efe-blackboard/src/blackboard.rs`) is a `DashMap` keyed by JSON pointers: each cell carries its value, a **monotonic per-pointer version**, the authoring agent and a timestamp. The conditional write `write_if` is a real compare-and-swap under a single shard guard (a post-audit atomicity fix). On top: global and per-namespace byte caps, a bounded ring of recent writes, and observability seams. The *EventBus* (`eventbus.rs`) is a bounded `tokio::broadcast` (capacity 256) publishing `SwarmEvents`.

The orchestrator (`efe-swarm-orchestrator/src/runner.rs`) allocates **one** fresh blackboard and **one** bus per run, then walks the DAG's layers: every non-skipped agent in a layer is spawned in parallel (`tokio::spawn`) with a per-agent timeout inside a global timeout, conditional edges and skip propagation. The agentic bridge (`efe-agentic-bridge/src/claude.rs`) calls the **real** Claude: a reqwest HTTPS client to `api.anthropic.com/v1/messages`, a two-turn *tool_use* protocol (one fixed tool `swarm_context`, the *ContextBundle* re-injected as the `tool_result`), SSE streaming, backoff retry, and fail-closed N17 provenance. It is a substantial, real Anthropic client.

> **Why "agentic blackboard = PROD" is only half true**
>
> Three decisive reservations, all verified in the code: (1) **Canned response by default** — without `ANTHROPIC_API_KEY`, the gateway selects `StubAgenticBridge`, and `/v1/agentic/call` returns "[stub bridge] would have called Claude…". The real Claude call is real-but-off-by-default. (2) **An essentially empty ContextBundle** — the served DAG never seeds `/input/text` (`seed_input_text` is only called in tests), and the aggregator is a `NoOpAggregator`: the analysis agents read empty text and the bundle only recaps agent ids and a cost. (3) **An EventBus with no subscriber** at runtime — every publish falls back to a blackboard write. The concurrency machinery genuinely runs; the analytic payload is near-zero on the served path.

To be credited nonetheless: the concurrency substrate (per-field versioning, atomic CAS, caps, bounded ring) is real and audited; the orchestrator (layered parallel scheduling, timeouts, best-effort isolation) is real; the Claude client is a real Anthropic client (two turns, streaming, retry, provenance); the gateway enforces real JWT/tenant auth. The **capability** "pub/sub bus for inter-agent events" is accurate; what is missing is a **live subscriber** and input seeding on the served path.

8 What we built — the chronological thread (M1.a → 665ead0)

Our work on inter-agent exchange is the Couche 3 M1 arc: opening, hardening, then making a real efe-core participant *servable* (opt-in) as the meta-swarm's operator Φ. Seven commits, ~22 h, July 13–14, 2026. Every milestone was locked fail-closed; every over-sell, retracted.

Commit	Milestone	Delivered (and the gotcha)	Status
8fcb3dc	M1.a — participant seam	Opens the seam: the meta step calls a real `SwarmController` via `phi_once`; `step_with_pull` → coupled mix. *Gotcha*: Mode-1 has min_rounds=4/hysteresis → a "converge-now" detector was needed.	real, tested
720d07c	M1.b — deterministic Φ	`init_sigma=0` on the latent controller → bit-reproducible Φ. *Gotcha*: the real value is seed-invariance (genesis noise derives from the spec id, not the fresh SwarmId).	real, tested
a8f1798	M1.c — Lipschitz probe	Revives the H1 guard: adversarial estimator (discontinuity bisection), empirical certificate, strict gate. *Gotcha*: a *fail-open* bug (`.ok()` swallowed Φ blow-ups) → fixed via a 3-state `PhiOutcome`.	real, off
c53a7f6 + d3ec501	M1.d — participants SERVED	Wires the efe-core bridge into the gRPC layer (engine block, config_hash, boot gate, require_convergence). *Gotcha*: an over-generic .gitignore swallowed SOURCE dirs → a fresh clone did not build.	real, off
b9a8fce	Palier-2 — fixed topology	Removes the routing discontinuity (Φ continuous by construction). **The deciding measurement**: Ł 0.366 (fixed) vs 0.691 (dynamic) — the analytic certificate $\beta_1 \approx 0.17$ is refuted.	real, off
665ead0	Review-debt discharge	Closes the M1.d adversarial review (10 findings). #1 = a real soundness bug (engine-identity reload bypass); #2/#6/#8 = over-sell retractions (the confounded ~1.9 ratio, over-assertive docstrings).	fixed

Neighbouring-layer context: the COUCHE2 arc (decision + served surface) is earlier and distinct — it gave Layer 2 a re-verifiable receipt in efe-swarm-gateway, with the honest trust model "R4 = tamper-evident / key-holder-verifiable, NOT trustless-third-party".

Two honesty meta-facts about this chronology. First, **the served milestone preceded its review**: M1.d was committed under credit constraints, its adversarial review initially deferred, then discharged ~12 h later (665ead0) — including a real soundness bug. Second, the .gitignore fix (d3ec501) means the pushed repository **did not compile** from a fresh clone before mid-arc (the manifest module and efe-telemetry/logs were untracked).

9 The honesty ledger

This section is the report's credibility spine: for every *tempting* claim, the verified *reality* in the code. It is the filter to run before any Kovant-facing presentation.

Tempting claim	Reality, verified in the code
"The platform coordinates Claude/Gemini/GPT in production."	**False for the default build.** Served only in efe-swarm-gateway, feature `frontier-llm` (off) + flag + boot triad. efe-server does not depend on it.
"The meta-swarm runs real swarms/SLMs."	By default it is a **mock** Φ=κ·x. The real efe-core participant is off (`core-integration`), and these are **numerical latent swarms**, not SLMs (no weights, no tokenizer).

Tempting claim	Reality, verified in the code
“Weighted / robust / Byzantine aggregation between agents.”	**None.** Only `UniformMeta0p` (a mean, Lipschitz=1) is instantiated. One adversarial participant moves the consensus. `AttentionMeta0p` is a stub (can be expansive).
“Non-expansiveness proven ($L \le \beta_1 \approx 0.17$).”	**Refuted as a certificate.** Measured $\hat{L} \approx 0.37$ (fixed) / 0.69 (dynamic) $\gg 0.17$. The probe is a **falsifier** (a lower bound), never a proof.
“Fixing the topology divides the sensitivity by ~1.9.”	**Retracted (confounded).** Fixed = 4 agents, dynamic ≥ 128 (population ~32× + mix change with the routing). Only “the fixed operator alone: $0.366 \gg 0.17$” survives.
“Non-expansiveness certificates exchanged on the wire.”	**Stub.** `verify_non_expansive` always returns Stub; H1 is only checked at **boot**. Signed certificates (Ed25519) are milestone M4, not built.
“The agentic blackboard coordinates real LLM agents.”	**Canned response by default** (no key). Moreover the served DAG does not seed the input → a near-empty ContextBundle; the EventBus has no runtime subscriber.
“The swarm self-diversifies / recovers from collapse.”	**No-op stubs.** `respawn_diversify` and `inject_noise` log and return Ok; the Mode-1 wiring disables recovery.
“The four layers coordinate through a shared blackboard.”	**False.** Four independent binaries, no shared data path at runtime (only a metrics *type* is reused).

What is genuinely solid must also be **credited**, because understating is as dishonest as inflating: Layer 1’s latent fusion (served, deterministic, tested); the Lipschitz probe (a real adversarial falsifier that caught its own fail-open bug in review); the fail-closed gates (boot H1, require_convergence→ABORTED, the kappa/seed Option asymmetry, the engine-identity reload guard); the mTLS transport; and Layer 2’s re-verifiable provenance receipt. The **honesty machinery** is real, even where the guarantees are downgraded.

10 The current session — discharging the review debt (665ead0)

The latest commit is the one we just produced: M1.d’s deferred adversarial review, its 10 findings fixed, re-verified, then re-screened by a four-lens adversarial review (0 confirmed findings).

The substantive fix (#1) — the engine-identity reload bypass

The reload check only compared ids present in **both** manifests: an id **removed** then **re-added** with a different engine, across two SIGHUP reloads, escaped the drift check (AD5 allows add/remove). We added an engine-identity **lineage guard** (`validate_engine_lineage` + an `engine_lineage` registry seeded at boot, monotonic “first identity wins”), factoring out the shared comparator `participant_engine_drift`. Honesty note: the **participant side** was already the sound backstop (it compares against *running* state, immune); this only fixes the divergence of the controller’s *book*.

The over-sell retractions and the test teeth

- **#2/#6/#8** — retraction of the confounded causal “~1.9” claim; docstrings aligned with the measurement ($0.37 > \beta_1$, sensitivity beyond *beta_mix*); the `ControllerManifest` doc fixed.
- **#3/#4/#5** — the boot gate’s REFUSE path (impossible margin), an anti-vacuous contraction guard (`first > 1e-4`), a **non-causal** fixed-vs-dynamic distinguishing assertion (`|Δ| > 0.1`).
- **#7/#9/#10** — the StepOnce exemption documented + pinned; the participant reload guard (step 2bis) tested.

Verification (before commit)

Default suite: **20 binaries, 0 failures** (lib 368). Feature suite `core-integration`: **27 binaries, 0 failures** (efe-core e2e over gRPC 5/5, manifest-engine 7/7, require_convergence 4/4). Four-lens adversarial re-review (~395 k tokens): **0 confirmed findings**. Commit `665ead0`, pushed to `couche2-m0-frontier`.

11 What remains — an honest roadmap

Making inter-agent exchange genuinely "production-grade" means closing, in order of value, the gaps this report has named without masking them:

- **Serve the real participant by default** — take efe-core out of `core-integration` off, which requires wiring certificates **on the wire** (milestone M4) so strict mode does not brick the served path (today the guarantee goes through the `ABORTED` path, not certificates).
- **Real robust aggregation** — at least a served median/trimmed scheme or outlier rejection, so the consensus survives an adversarial participant (today: a uniform mean).
- **Reconcile the $\beta_1 \approx 0.17$ vs $\mathcal{L} \approx 0.37$ gap** — attribute the residual sensitivity through stage-by-stage measurement (via `phi_once` on assembly variants); re-measure fixed-vs-dynamic **on a matched population** to lift the confound.
- **Consolidate mode1** — merge the fixed-topology controller (a duplicated server-side mirror) into a `build_mvp_controller_fixed` (acknowledged debt, drift risk); and land the uncommitted "pure Φ" fix cleanly.
- **Layer 2** — fully apply `call_many_ok → call_many` (vote), structurally deduplicate the egress guard, and — if desired — wire the frontier surface into the product server.
- **Layer 4** — seed `/input/text` on the served path, replace the `NoOpAggregator`, and attach a real EventBus subscriber, without which the agentic blackboard stays an empty shell by default.

12 Annex — index of crates, binaries and milestones

Binaries serving an exchange layer

Binary	Layer	Exchange served by default	Key crates
efe-server	1	Mode-1 latent fusion (SERVED)	efe-core, efe-server/mode1.rs
efe-swarm-gateway	2	frontier panel (OFF: feature+flag)	efe-coordinator, efe-multi-llm, provenance-capture
efe-meta-swarm-server	3	gRPC meta-swarm (MOCK κ·x)	efe-meta-swarm, efe-meta-swarm-server
efe-gateway	4	agentic blackboard (CANNED without a key)	efe-blackboard, efe-swarm-orchestrator, efe-agentic-bridge

Test footprint (static grep counts — not re-run, unless stated)

- **Layer 1** (efe-core fusion/genesis/els/dissolution/swarm): ≈ 427 inline tests (grep).
- **Layer 2**: ≈ 131 test fns (efe-multi-llm), ≈ 727 attrs (efe-coordinator), ≈ 36 (gateway llm.rs) — grep.
- **Layer 3**: the M1 suites **re-run this session** — default 368 lib + 20 bins, feature 27 bins, 0 failures.

- **Layer 4**: ≈ 675 test attrs across the 5 crates (grep; `audit_*` / `chaos_*` suites + Anthropic mocked via wiremock).

> **Method note & final reservation**
>
> This report is grounded on a multi-agent code cartography (file:line citations) and a dedicated honesty review. The test counts for layers 1, 2 and 4 are **static grep estimates**, not verified green runs; only the M1 suites (Layer 3) were re-run this session. No multi-binary production deployment is established here: the existence of four servers does not prove a deployed topology. When in doubt, this document always chooses the lower formulation — per the cardinal rule.